

# ViT5 as 1-best Error Correction model for Vietnamese spoken question answering

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## 1. Research Context and Hypothesis

- Current study's aim is to research the effectiveness of using language models as error correction (EC) models to correct Automatic Speech Recognition (ASR) induced errors in spoken question answering (SQA) in Vietnamese.
- SQA in Vietnamese remained underrepresented [2] compared to textual QA [2] [3]. The ViSQA dataset (Vietnamese Spoken Question Answering) [2] emerged to fill that gap. Its study noted:
  - ASR-induced errors, such as deletions, substitutions, and misalignments significantly impact QA model effectiveness
  - Enhancing ASR-fidelity through semantic-preserving error correction targeting entities, numbers or temporal markers directly strengthen SQA robustness
- Previous study studied language models as EC model used a supervised approach with N-best list to train a T5 language model [1], which this study aims to replicate but with a simple 1-best output from the ASR system instead of the N-best list.
  - RQ1:** How well does a language model (ViT5) work as an error correction/post-processing step for aligning Vietnamese ASR transcriptions to the ground truth?
    - Hypothesis** – Based on previous work, using a language model can work to align Vietnamese ASR transcription to ground truth in terms of Word Error Rate (WER), even with a 1-best hypothesis approach as opposed to N-best hypotheses
  - RQ2:** Does ASR post-processing error correction model improve downstream Vietnamese spoken question-answering performance compared with using the original ASR transcription?
    - Initial Hypothesis** - Adding the error correction model will improve Vietnamese spoken question answering performance compared to using the raw ASR transcription
    - After RQ1 was completed, hypothesis changed** – introduction of hallucinations to hypotheses might lower the question answering performance in terms of Exact Match (EM) and F1

## 2. Methodology

- Research evaluated on ViSQA dataset, and used both audio files and transcription generated by AssemblyAI [2] of the dataset
  - Audio files were ran through PhoASR [10] to create a train and a second test set
- Because ViSQA was based on UiT-ViQUAD 2.0 dataset [2], the latter dataset was used in this experiment as ground truth context and answers

### 2.1. Research Question 1

- 3 conditions were experimented: no EC model; Base ViT5 EC Model [5]; and Fine-tuned ViT5 EC Model
  - Fine-tuned ViT5 EC Model was done by training the model to correct PhoASR train set contexts to corresponding UiT-ViQUAD 2.0 train set [9] ground truth contexts (black arrow in Figure 1)

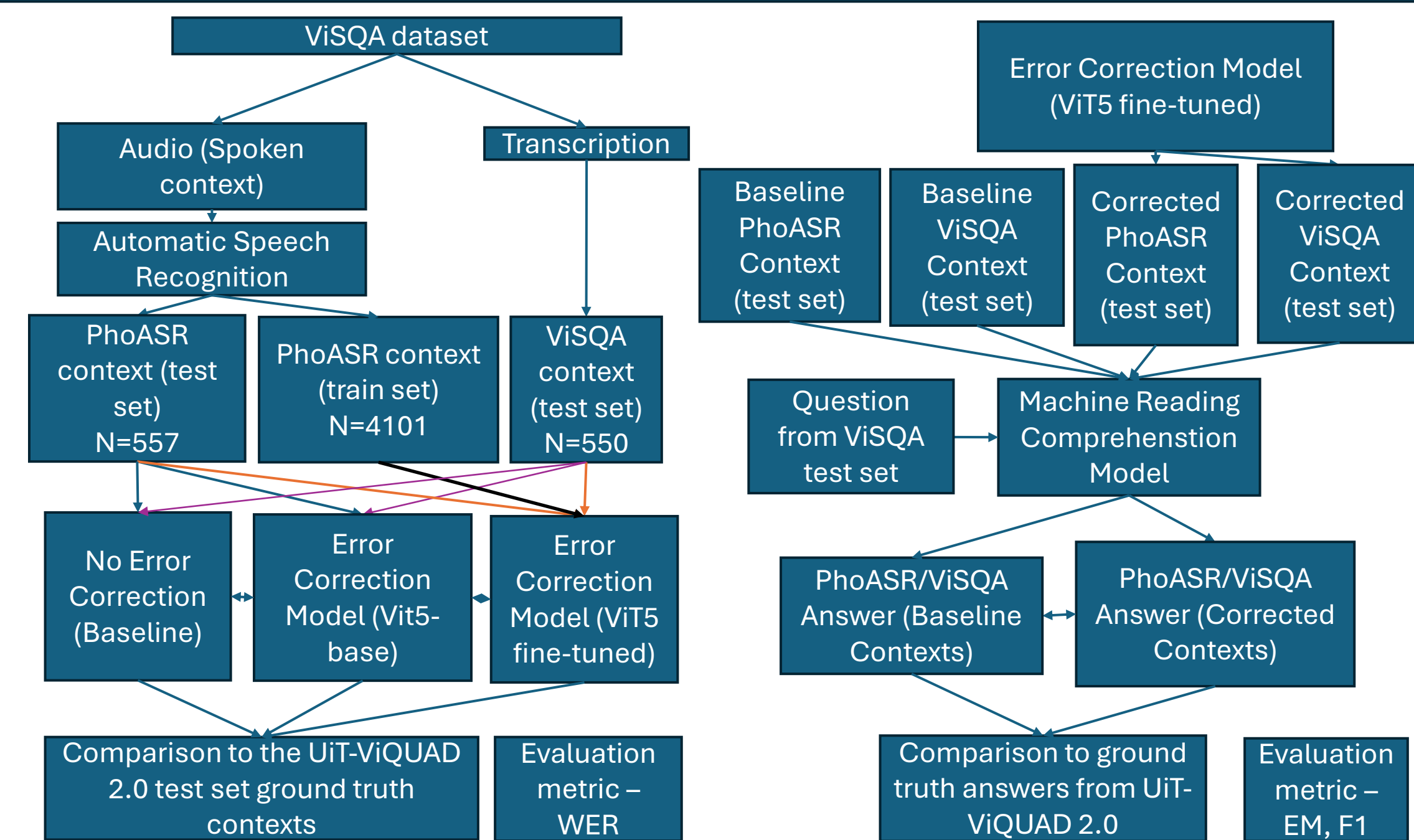


Figure 1: Overview diagram of the methodology for RQ1 (arrow colors indicate which set was used where)

### 2.2. Research Question 2

- The fine-tuned EC model from RQ1 takes both PhoASR-generated and ViSQA contexts as input for correction (orange arrows in Figure 1). The output corrected contexts, along with their original contexts were put through QA models for Machine Reading Comprehension tasks using questions from ViSQA test set
- Answers from different contexts were compared to ground-truth answers
- QA Models used was based on previous literature [2] (these are public Hugging Face checkpoints fine-tuned with UiT-ViQUAD 2.0 dataset):
  - heidie/phobert\_finetuned\_viquad [4]
  - PhucDanh/vit5-fine-tuning-for-question-answering [7]
  - PhucDanh/Bartpho-fine-tuning-model-for-question-answering [6]
  - NamSyntax/xlmr-large-viquad [8]

## 3. Experimental Evaluation & Results

| Mean ASR WER (No Error Correction) |        | Mean Base ViT5 WER (Base Error Correction) |        | Mean Fine-tuned ViT5 WER |        |
|------------------------------------|--------|--------------------------------------------|--------|--------------------------|--------|
| ViSQA                              | PhoASR | ViSQA                                      | PhoASR | ViSQA                    | PhoASR |
| 0.32                               | 0.31   | 1.59                                       | 1.99   | 0.17                     | 0.19   |

Table 1: Transcription accuracy compared to UiT-ViQUAD 2.0 test set ground truth contexts across 3 conditions

### 3.1. Research Question 1

- WER decreased in the third condition, findings in accordance with original hypothesis**
- It is unexpected that the base ViT5 model without any fine-tuning would result in a much higher WER compared to the raw ASR transcripts
- By comparison, fine-tuning the model by having a **clear input to target training set** improves the alignment of raw ASR transcript to ground truth substantially

However, in the latter two conditions, **ViT5 introduced hallucinations to the ASR transcript**; most likely with regards to named entities (however the exact numbers of this were not quantified in this study)

- For a QA system, this is unacceptable, as it actively changes the semantics of the transcripts [2], **thus leading to the hypothesis change for RQ2**

| Model   | ViSQA-test [2] |        | ViSQA-test baseline context |        | ViSQA-test corrected context |        |
|---------|----------------|--------|-----------------------------|--------|------------------------------|--------|
|         | EM (%)         | F1 (%) | EM (%)                      | F1 (%) | EM (%)                       | F1 (%) |
| PhoBERT | 31.85          | 53.32  | 0.08                        | 31.58  | 0.23                         | 32.81  |
| ViT5    | 36.30          | 63.46  | 26.21                       | 54.74  | 29.59                        | 58.49  |
| BartPho | 23.42          | 53.17  | 26.74                       | 55.47  | 30.35                        | 61.15  |
| XLMR    | 33.05          | 55.98  | 29.67                       | 57.83  | 32.63                        | 62.13  |

Table 2: Model evaluation of the effects of ViSQA answers compared to UiT-ViQUAD 2.0 test set ground truth answers across 2 context conditions. The left-most section are the results taken from source material, which also compared the answers from the models with the UiT-ViQUAD ground truth answers

| Model   | PhoASR-test baseline context |        | PhoASR-test corrected context |        |
|---------|------------------------------|--------|-------------------------------|--------|
|         | EM (%)                       | F1 (%) | EM (%)                        | F1 (%) |
| PhoBERT | 0.15                         | 33.90  | 0.30                          | 33.20  |
| ViT5    | 20.62                        | 49.29  | 28.72                         | 57.02  |
| BartPho | 21.41                        | 50.99  | 29.74                         | 59.61  |
| XLMR    | 24.20                        | 52.86  | 32                            | 60.60  |

Table 3: Model evaluation of the effects of PhoASR answers compared to UiT-ViQUAD 2.0 test set ground truth answers across 2 context conditions.

### 3.1. Research Question 2

- Despite the introduction of possible hallucinations regarding entities, **EM and F1 in general increased across all models, initial hypothesis for RQ2 held**
  - Could mean that the effect that hallucinations observed in the corrected contexts are not as high as expected

## 4. Limitations, Discussion, and Conclusions

### 4.1. Limitations

- ViT5 EC model observed to hallucinate, especially on named entities. Whereas the ASR system may recognise the entity strangely, putting the transcript through a LM may change it to a different named entity
- Anomalous performance by the fine-tuned PhoBERT model compared to other models, no explanation can be found to the behavior so far
- ViSQA's audio files was created by using Chirp HD3 Google Text-to-speech system [2], not actual Vietnamese speakers

### 4.2. Discussion

- While QA performance improvement was observed, using the n-best list approach by [1] may yield better results insofar as using the LM as an EC model through supervised learning
  - The n-best list approach was unable to be implemented in this experiment, as PhoASR during preliminary trials generated only a single hypothesis for all trial audio files, even with a larger beam size and/or temperature

### 4.3. Conclusion

Using a language model as a 1-best EC model through supervised learning can help align Vietnamese ASR transcription to ground truth; adding the post-processing step improves downstream SQA performance. However, it introduces semantic hallucinations, and further work in the field of SQA should focus on mitigating that effect.